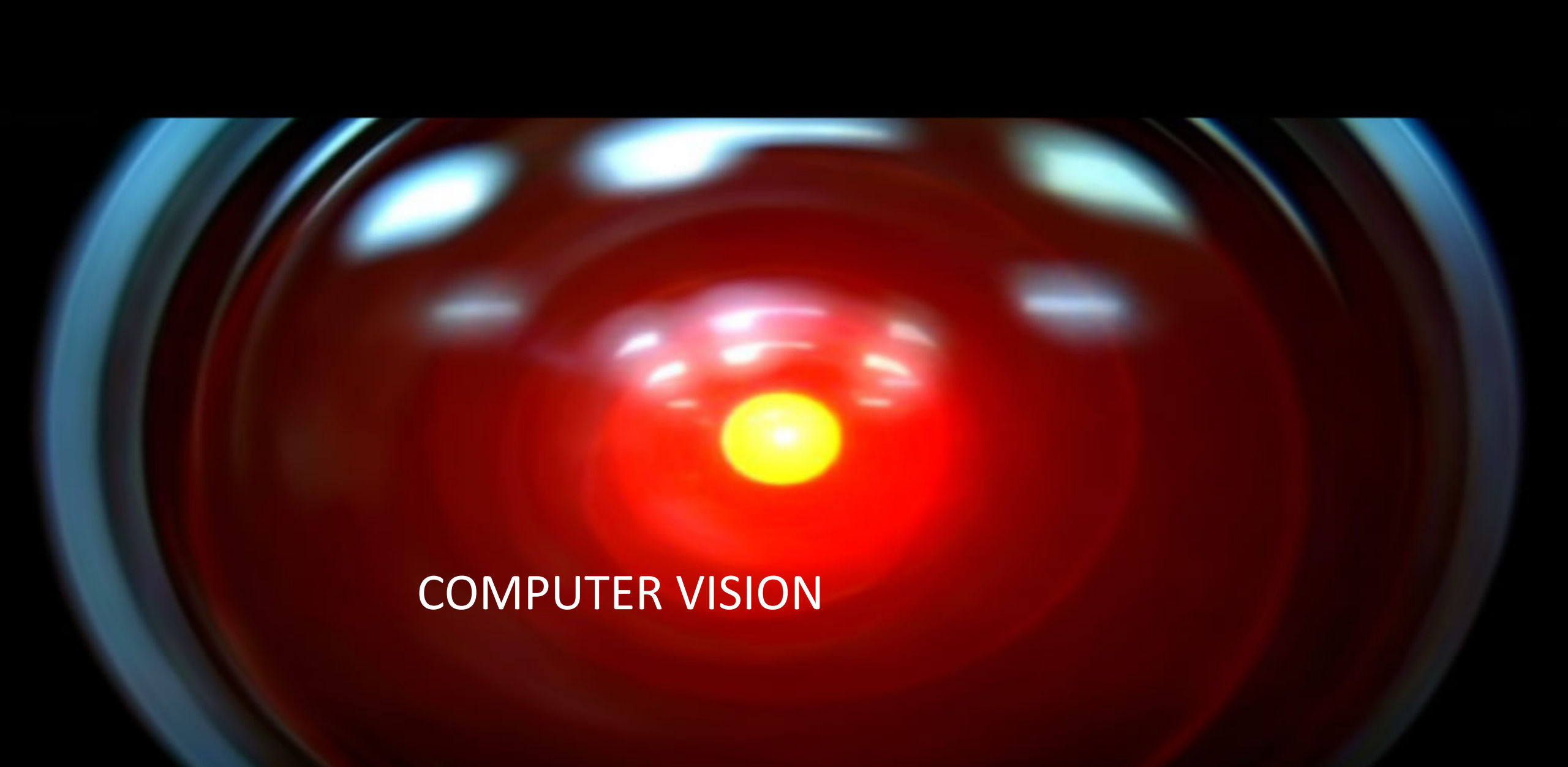




COMPUTER VISION

MATT/MET CVDL

Visual Recognition

Unit 4

Visual recognition : LAB2

Verónica Vilaplana

Visual Recognition Lab2a & Lab2b

- The goal of this Lab is to get hands on experience designing and training deep convolutional networks for image classification using PyTorch.
- In this assignment you will use PyTorch
- The assignment is divided in two parts. You will need to complete **just one part**, depending on your previous background on Deep Learning
 - **Lab2a** is mandatory for students that have not done DLAI in a previous semester
 - **Lab2b** is mandatory for students that have already done DLAI
- **Lab2a:** you will train and test a simple CNN to classify images from the CIFAR10 dataset
- **Lab2b:** you will learn how to train your network using transfer learning. The two major transfer learning scenarios are: (a) Using the CNN as fixed feature extractor, and (b) finetuning the CNN. You will use a model pre-trained on ImageNet for classifying the Caltech-UCSD Birds dataset (200 categories of birds, 3000 images in train and 3033 images in test).

Summarize Visual Recognition labs

- Write a report summarizing your work and results obtained in Lab1 and Lab2
- Use any 2-column template in word or latex
- Upload a pdf file
- You can use ChatGPT or similar tools to improve the quality of your report, but if you use it, you have to mention it explicitly.

Recommended papers

- Finally, this is not a mandatory task but if you are interested in visual recognition I strongly recommend you to read the following papers:

Segment Anything

Alexander Kirillov^{1,2,4} Eric Mintun² Nikhila Ravi^{1,2} Hanzi Mao² Chloe Rolland³ Laura Gustafson³
Tete Xiao³ Spencer Whitehead Alexander C. Berg Wan-Yen Lo Piotr Dollár⁴ Ross Girshick⁴
¹project lead ²joint first author ³equal contribution ⁴directional lead
Meta AI Research, FAIR

Segment Everything Everywhere All at Once

Xueyan Zou^{*§2}, Jianwei Yang^{*‡1}, Hao Zhang^{*‡}, Feng Li^{*‡}, Linjie Li[†], Jianfeng Gao^{¶‡}, Yong Jae Lee^{¶§}
[§] University of Wisconsin-Madison [‡] Microsoft Research at Redmond [‡] HKUST [†] Microsoft Cloud & AI
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DINOv2: Learning Robust Visual Features without Supervision

Maxime Oquab^{**}, Timothée Darcet^{**}, Théo Moutakanni^{**},
Huy Vo^{*}, Marc Szafraniec^{*}, Vasil Khalidov^{*}, Pierre Fernandez, Daniel Haziza,
Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba,
Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat,
Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal¹,
Patrick Labatut^{*}, Armand Joulin^{*}, Piotr Bojanowski^{*}

Meta AI Research ¹Inria
^{*}core team ^{**}equal contribution